

Group-Structured Attributions for Explainable Quantum Machine Learning

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Abstract—Explainable quantum machine learning needs methods that can identify not only important individual gates, but also important functional parts of a quantum circuit. This thesis adapts Owen values to quantum-circuit explanations by treating selected gates as players, grouping them into functional modules, and evaluating each coalition by running the circuit with only the selected gates kept active. The method is evaluated on a benchmark dataset of three-qubit circuits spanning different levels of magic and entanglement. In this benchmark, stabilizer Rényi entropy and Meyer–Wallach entanglement are used as quantum-resource value functions. The method is also evaluated on two quantum machine learning case studies, where classification accuracy is used as the value function to explain task performance. The benchmark results show that Owen values produce meaningful group-level explanations of quantum resources: the dominant groups align with the expected functional roles of the circuit modules. Group ablation further supports these rankings, since removing highly ranked groups generally causes larger changes in the target value than removing low-ranked groups. Since exact Owen computation becomes expensive as the number of gates increases, a Monte Carlo estimator is also evaluated. The estimator preserves the main Owen-value rankings while reducing exact enumeration cost. In the quantum machine learning case studies, Owen values partly retain Shapley-style gate-level structure while adding a clearer module-level interpretation. As a proof of concept, these results show that Owen values provide a useful complementary explanation framework for group-structured quantum circuits.

I. INTRODUCTION

Quantum Machine Learning (QML) is a fast-growing field at the intersection of machine learning and quantum computing [1], [2]. Many QML approaches use parameterized quantum circuits as part of machine-learning models [3]. As QML circuits become more complex, it becomes harder to understand which parts of the circuit are actually important. This motivates explainable quantum machine learning (XQML): methods that help us inspect quantum models and understand why they behave in a certain way [4], [5].

A recent approach called Shapley values for quantum circuit explanations (SVQX) explains quantum circuits by treating gates as players in a cooperative game [5]. Each gate receives a Shapley value, which measures how much that gate contributes to a chosen value function, such as classification accuracy. This

builds on the broader use of Shapley values in explainable artificial intelligence [6].

However, quantum circuits are often not best understood as lists of separate gates. They are usually designed from larger functional parts, where different gates or groups of gates contribute to different circuit functions [7]. For example, some parts of a circuit may mainly contribute to magic, while others may contribute to entanglement, and these resources can interact within the same circuit [8]. A purely gate-level explanation may therefore be too detailed to show which larger circuit mechanism is responsible for a result.

This thesis investigates whether Owen values can address this limitation. Owen values extend Shapley values to games where players are divided into predefined groups [9], [10]. We propose an Owen-value framework that produces attributions at both the gate level and the functional-group level, evaluated on a controlled benchmark and two QML case studies.

These gaps motivate the following research questions:

RQ1: *How can Owen values be adapted and computed, exactly and approximately, to explain the importance of individual gates and functional groups in group-structured quantum circuits?*

RQ2: *To what extent do Owen values produce meaningful group-wise explanations on a controlled quantum-circuit benchmark, and are these explanations supported by group-ablation tests?*

RQ3: *How do Owen-value explanations compare with standard gate-level Shapley-value explanations in QML case studies, and what additional module-level interpretations do Owen values provide?*

II. BACKGROUND

A. Shapley and Owen Values

A cooperative game is defined by a set of players $N = \{1, \dots, n\}$ and a value function $v : 2^N \rightarrow \mathbb{R}$. The value $v(S)$ describes the score obtained by a coalition $S \subseteq N$. The Shapley value of player i is its average marginal contribution over all coalitions not containing i [11]:

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(n - |S| - 1)!}{n!} [v(S \cup \{i\}) - v(S)]. \quad (1)$$

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The Shapley value is used in explainable artificial intelligence to divide a model output among features or model components [6]. In quantum-circuit explanations, the players can be gates and $v(S)$ can measure the output of the reduced circuit that keeps only the gates in S [5].

Unlike Shapley values, Owen values incorporate a pre-defined group structure over the players [9], [10]. If $\mathcal{P} = \{G_1, \dots, G_m\}$ is a partition of N into groups, then $G(i)$ denotes the group containing player i . For player i , an Owen context is defined by two coalitions: an outer coalition $R \subseteq \mathcal{P} \setminus \{G(i)\}$ of groups, and an inner coalition $T \subseteq G(i) \setminus \{i\}$ of players inside the group of i . The union of the groups in R is

$$U_R = \bigcup_{G \in R} G. \quad (2)$$

The marginal contribution of player i in this Owen context is

$$\Delta_i(R, T) = v(U_R \cup T \cup \{i\}) - v(U_R \cup T). \quad (3)$$

The Owen value of player i is the weighted average of these marginal contributions:

$$\psi_i(v, \mathcal{P}) = \sum_{R \subseteq \mathcal{P} \setminus \{G(i)\}} \sum_{T \subseteq G(i) \setminus \{i\}} \alpha(R, T) \Delta_i(R, T), \quad (4)$$

where

$$\alpha(R, T) = \frac{|R|!(m - |R| - 1)!}{m!} \frac{|T|!(|G(i)| - |T| - 1)!}{|G(i)|!}. \quad (5)$$

Owen values thus measure a player's importance while respecting the chosen group structure. In this thesis, group-level importance is obtained by summing the Owen values of all active gates in the same group:

$$\Psi_G = \sum_{i \in G} \psi_i(v, \mathcal{P}). \quad (6)$$

B. Owen Values Under Uncertainty

The Owen value formula is deterministic once the value function is known. In quantum-circuit experiments, however, reduced-circuit values may be observed through finite measurement shots, noisy hardware, or stochastic model predictions. The observed coalition value can therefore be a noisy estimate of the underlying value, following the uncertainty view used in SVQX [5].

To reduce this uncertainty, the same coalition can be evaluated K times and averaged:

$$\bar{v}_K(S) = \frac{1}{K} \sum_{k=1}^K \tilde{v}^{(k)}(S). \quad (7)$$

Here, $\tilde{v}^{(k)}(S)$ is the observed value of coalition S in the k -th evaluation, and K is the number of repeated evaluations of the same coalition. Larger K usually gives a more stable coalition value, but also increases runtime. Using these averaged values, the marginal contribution of gate i in an Owen context is estimated as

$$\hat{\Delta}_{i,K}(R, T) = \bar{v}_K(U_R \cup T \cup \{i\}) - \bar{v}_K(U_R \cup T), \quad (8)$$

where R is a coalition of groups, T is a coalition inside the group of gate i , and U_R is the union of the groups in R .

C. Value Functions

A value function defines what property or behaviour is being explained. In this thesis, two types of value functions are used. Property-based value functions, such as magic and entanglement, assign a score to a physical property of the output quantum state, while task-based value functions assign a score to model performance, such as classification accuracy.

For a circuit C on q qubits, the output state is evaluated on the standard all-zero input state:

$$|\psi_C\rangle = C|0\rangle^{\otimes q}. \quad (9)$$

The first property is magic, measured by stabilizer Rényi entropy (SRE) [12], [13]. Magic is a quantum resource associated with states outside the stabilizer/Clifford setting, enabling universal quantum computation when combined with Clifford operations [14]. For a q -qubit pure state $|\psi\rangle$, the stabilizer 2-Rényi entropy used in this thesis is

$$M_2(|\psi\rangle) = -\log \left(\frac{1}{2^q} \sum_{P \in \mathcal{P}_q} \langle \psi | P | \psi \rangle^4 \right), \quad (10)$$

where $\mathcal{P}_q$ is the set of q -qubit Pauli strings. The logarithm is the natural logarithm. Stabilizer states have zero SRE, while larger values indicate more magic. The corresponding circuit-level score is

$$f_{\text{magic}}(C) = M_2(|\psi_C\rangle). \quad (11)$$

The second property is entanglement. This thesis uses the Meyer–Wallach global entanglement measure [15], following Brennen's formulation for pure multi-qubit states [16]. For a q -qubit pure state $|\psi\rangle$, ρ_k is the reduced density matrix of qubit k . The Meyer–Wallach measure is

$$Q(|\psi\rangle) = 2 \left(1 - \frac{1}{q} \sum_{k=1}^q \text{Tr}(\rho_k^2) \right). \quad (12)$$

The corresponding circuit-level score is

$$f_{\text{ent}}(C) = Q(|\psi_C\rangle). \quad (13)$$

In this thesis, the property-based scores are deterministic because they are computed using noiseless statevector simulation. The full output state is obtained directly instead of being estimated from finite measurement shots.

For the QML case studies, the task-based score is classification accuracy. For a test set $\mathcal{D}_{\text{test}} = \{(x_j, y_j)\}_{j=1}^{d_{\text{test}}}$, the accuracy of a model is

$$\text{Acc} = \frac{1}{d_{\text{test}}} \sum_{j=1}^{d_{\text{test}}} \mathbf{1}[\hat{y}_j = y_j], \quad (14)$$

where $\hat{y}_j$ is the predicted label for input x_j .

Thus, the Owen-value framework is not tied to one specific value function: SRE and Meyer–Wallach entanglement are used to explain quantum resources, while accuracy is used to explain QML model performance [5].

III. METHODS

This thesis extends the SVQX reduced-circuit idea with Owen values. The Owen-value framework is applied to a controlled 35-circuit magic–entanglement benchmark, in which active gates form the player set, circuit modules define the coalition structure, and property-based value functions define what is being explained.

A. Group-Structured Circuit Explanations

In this thesis, the cooperative-game players are selected active gates, while gates not selected for attribution are passive and kept fixed in every coalition evaluation. For a coalition $S \subseteq N$, the reduced circuit C_S contains all passive gates and only the active gates in S , preserving the original gate order. For a circuit property or task score f , the coalition value is

$$v_f(S) = f(C_S). \quad (15)$$

For the benchmark experiments, this gives

$$v_{\text{magic}}(S) = f_{\text{magic}}(C_S) \quad \text{and} \quad v_{\text{ent}}(S) = f_{\text{ent}}(C_S). \quad (16)$$

For the QML case studies, $v_{\text{acc}}(S)$ is the classification accuracy of the reduced model induced by coalition S .

The active gates are further divided into predefined groups. Owen values are computed over this grouped player set, giving gate-level attributions, while group-level scores are obtained by summing the Owen values of all active gates in the same group using Eq. (6).

B. Coalition Structure

The main coalition structure used in this thesis is a resource-based E/M/X partition. Active gates are divided into three groups:

- *E*: entanglement-related gates, mainly CNOT/CZ gates;
- *M*: local magic-related gates, mainly non-Clifford or parameterized single-qubit gates;
- *X*: mixed gates, mainly non-Clifford or parameterized gates that are embedded in, or directly adjacent to, entangling motifs.

Gates outside these active roles are treated as passive. The partition is manually designed from physical intuition, with groups corresponding to entanglement generation, local magic generation, and mixed local–entangling effects.

The E/M/X partition is used for the 35-circuit benchmark and as the main resource-based partition in the QML case studies. A label-aware random-partition control on the exact benchmark subset is reported in Appendix A-B to test whether this physically motivated grouping aligns with the expected resource roles more strongly than matched random groupings. The full benchmark partitions with the circuit structures are reported in Appendix A-A.

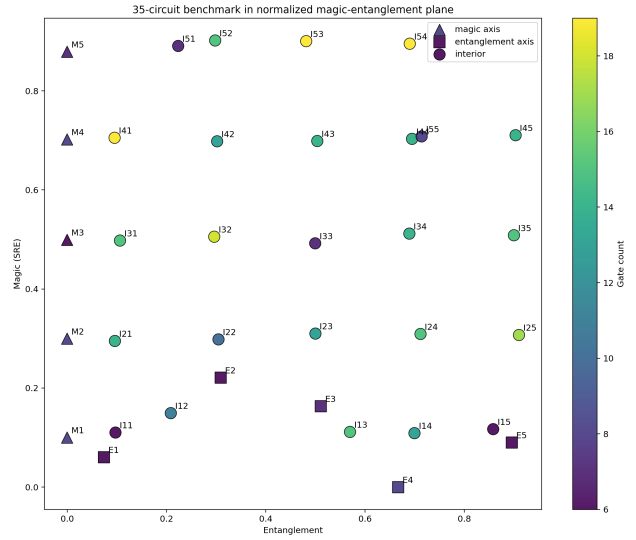


Fig. 1. Controlled benchmark of 35 three-qubit circuits in the normalized magic–entanglement plane.

C. Controlled Benchmark Construction

A controlled benchmark is used to assess whether Owen values can separate magic-related and entanglement-related circuit contributions. A pool of 10,000 random three-qubit circuits was generated as a large but computationally manageable search space for selecting circuits with different magic and entanglement levels. The circuit-generation and SRE-labelling code was adapted from the repository accompanying Lipardi et al. [13]. For each circuit, the number of gates was sampled uniformly from 6 to 19. One of three gate families was selected with equal probability: rotations with CNOT gates, Clifford-like gates with rotations and CNOT gates, or Clifford+*T* gates. Rotation angles were sampled uniformly from $[0, 2\pi)$.

Each circuit was labelled using the two property-based value functions from Section II-C: SRE for magic and the Meyer–Wallach measure for entanglement. Both values were normalized over the full pool. From this labelled pool, 35 circuits were selected to cover the normalized magic–entanglement plane, as shown in Fig. 1. The benchmark was designed to contain five magic-axis circuits with low entanglement and increasing magic, five entanglement-axis circuits with low magic and increasing entanglement, and 25 interior circuits with mixed values. Because the benchmark is selected from a finite random pool of small three-qubit circuits, the axes are approximate rather than perfectly ideal. In particular, the entanglement-axis circuits were chosen to have low magic while increasing in entanglement, but they do not all have exactly zero SRE. This reflects the fact that, for the available gate families, circuit depths and number of qubits, magic and entanglement cannot always be controlled independently.

The 15 shortest circuits are used for exact Owen computation, where all Owen contexts can still be enumerated. The full set of 35 circuits is used for the sampled Owen study.

All benchmark value functions are computed with noiseless

statevector simulation. Thus, the SRE and Meyer–Wallach values are deterministic and are not estimated from measurement shots.

D. Exact and Sampled Owen Computation

Owen values can be computed exactly when the number of active gates and groups is small enough. Exact computation means that all Owen contexts are evaluated: for each active gate, all outer coalitions of groups and all inner coalitions inside the gate’s own group are enumerated. In this thesis, exact Owen computation is used for the 15 short benchmark circuits and for the QML case studies.

For larger circuits, exact enumeration becomes expensive. A Monte Carlo Owen estimator is therefore introduced for the full 35-circuit benchmark, adapting the coalitional-value estimator of Kotsiopoulos et al. [17] to sample only Owen contexts (R, T) , since the value function is evaluated directly for each context. For gate i , L contexts (R_ℓ, T_ℓ) are sampled with probabilities proportional to the Owen weights $\alpha(R, T)$ in Eq. (5), giving

$$\hat{\psi}_i = \frac{1}{L} \sum_{\ell=1}^L [v(U_{R_\ell} \cup T_\ell \cup \{i\}) - v(U_{R_\ell} \cup T_\ell)], \quad (17)$$

where U_{R_ℓ} is the union of the groups in R_ℓ . The weights do not appear explicitly in Eq. (17) because they are used as the sampling probabilities.

For stochastic value functions, each coalition value in Eq. (17) is replaced by its averaged estimate $\bar{v}_K$ from Eq. (7). For deterministic value functions $K = 1$ is sufficient.

The estimator is validated on the exact 15-circuit subset using sampling fractions 0.30, 0.50, 0.70, and 0.90 with 10 random seeds per circuit and value function. Here, a sampling fraction refers to the fraction of the available Owen context space sampled for each gate. For example, a sampling fraction of 0.70 means that the number of Monte Carlo samples for that gate is set to approximately 70% of the number of exact outer-inner Owen contexts. The sampled Owen values are compared with exact Owen values using maximum absolute error, normalized error, Spearman rank correlation, and runtime. For an active-gate set N , the normalized error is defined as

$$\text{NE} = \frac{\frac{1}{|N|} \sum_{i \in N} |\hat{\psi}_i - \psi_i|}{\sum_{i \in N} |\psi_i| + 10^{-12}}. \quad (18)$$

Here, $\hat{\psi}_i$ is the sampled Owen value and ψ_i is the exact Owen value of gate i . Thus, the normalized error is the mean absolute difference between sampled and exact gate-level Owen values, divided by the total absolute magnitude of the exact gate-level Owen-value vector. This makes the error relative to the scale of the exact attribution vector. Based on this validation, the full 35-circuit study uses sampling fraction 0.70 and averages five sampled runs per circuit and value function.

E. Meaningfulness and Group-Ablation Criteria

The benchmark evaluates meaningfulness and ablation-based support for the Owen-value rankings. Meaningfulness

is assessed by comparing the dominant Owen groups with the expected circuit roles. The dominant group is defined as the group with the largest signed group-level Owen total Ψ_G . For the magic value function, M or X is expected to be important because these groups contain non-Clifford or magic-related gates. For the entanglement value function, E is expected to be important because it contains the main entangling gates. Interior circuits are expected to show more mixed attributions.

Ablation-based ranking support is evaluated using group removal. For each circuit and value function, the highest-ranked and lowest-ranked Owen groups are removed separately from the full circuit, and the value function is recomputed. For a group G , the value drop is

$$\Delta_G = v(N) - v(N \setminus G). \quad (19)$$

A ranking supported by ablation should usually produce a larger drop for the top-ranked group than for the bottom-ranked group. This is a diagnostic check rather than a proof, because near-zero values and tied group attributions can make the comparison ambiguous.

IV. EXPERIMENTS

The experiments apply the Owen-value framework to two SVQX QML classification case studies [5]: a stochastic one-shot QNN accuracy setting and a deterministic kernel-based QSVM setting. Together, they test whether Owen values add useful module-level interpretations relative to gate-level Shapley/SVQX explanations.

A. Quantum Neural Network Case Study

The first QML case study uses a quantum neural network (QNN). A QNN is a classifier in which a parameterized quantum circuit processes an input and a measured qubit is used to predict the class label [3]. This case study follows the two-qubit QNN experiment from SVQX [5]. The circuit and gate indexing are reported in Appendix A-D. The trained parameters are kept fixed at $\theta \approx (3.860, -1.070, -1.583, 0.860)$ during the explanation experiment, and the circuit is evaluated on the same 20-point binary classification dataset [5]. The main E/M/X coalition structure for this QNN is reported in Appendix A-D.

The value function is classification accuracy, as defined in Eq. (14). The QNN circuit is evaluated with Qiskit Aer `qasm_simulator` using one shot per input. Therefore, each prediction is obtained from a single simulated measurement outcome, making the accuracy value function stochastic. To reduce this uncertainty, each coalition value is averaged over $K \in \{1, 8, 16, 32\}$ repeated evaluations, following Section II-B. Each setting is repeated over five independent runs. The resulting Owen explanations are compared with a gate-level Shapley baseline recomputed under the same active/passive convention.

B. Quantum Support Vector Machine Case Study

The second QML case study applies Owen values to the quantum support vector machine (QSVM) setup from SVQX

TABLE I
SUMMARY CHECKS FOR EXACT OWEN VALUES ON THE 15-CIRCUIT SUBSET.

Check	Result
Magic-axis circuits with top magic group in $\{M, X\}$	5/5
Magic-axis circuits with near-zero entanglement attribution	5/5
Entanglement-axis circuits with strict top group E	3/5
Entanglement-axis circuits where E is top or co-top	5/5
Interior circuits with shared or mixed attribution	5/5

[5]. A QSVM uses a quantum feature map to embed classical data into quantum states, then trains a classical support vector classifier on the resulting quantum kernel [18], [19]. The feature map is a two-qubit second-order Pauli-Z map with $r \in \{1, 2, 3\}$ repetitions; the gate indexing and E/M/X structure are reported in Appendix A-I. The phase functions are $\phi_1(x_1) = 2x_1$, $\phi_2(x_2) = 2x_2$, and $\phi(x) = 2(\pi - x_1)(\pi - x_2)$.

For each coalition S , the reduced feature map defines quantum states for all data points. Kernel entries are computed as squared state overlaps using deterministic statevector simulation: the main implementation uses a manual NumPy statevector simulator, with Qiskit’s `Statevector` simulator used as a reference check. A classical support vector classifier is then trained with the resulting precomputed kernel, and the coalition value is the test accuracy from Eq. (14).

Five train/test splits are used, each with 40 training points and 1000 test points. Since both the kernel evaluation and classifier training are deterministic, the QSVM value function gives the same result on repeated evaluations, so $K = 1$ is sufficient. Owen contexts are enumerated exactly for this case study.

V. RESULTS

This section reports the experimental results for the controlled benchmark, the Monte Carlo estimator validation, and the two QML case studies.

A. Exact Owen Values on the 15-Circuit Subset

The exact 15-circuit subset contains five magic-axis circuits, five entanglement-axis circuits, and five interior circuits. Table I summarizes the main group-level checks.

For the magic-axis circuits, the dominant magic group is in $\{M, X\}$ in all five cases, and the entanglement attribution is near zero in all five cases. For the entanglement-axis circuits, E is the strict top group in three out of five cases and is top or co-top in all five cases. The five interior circuits show shared or mixed attribution patterns.

B. Group-Ablation Support

Group ablation compares the value drop after removing the top-ranked Owen group with the value drop after removing the bottom-ranked Owen group. The results are shown in Fig. 2 and summarized in Table II.

For the magic value function, the mean top-group drop is 0.3502, compared with 0.0936 for the bottom group, and the top group causes the larger drop in 13 out of 15 circuits. The

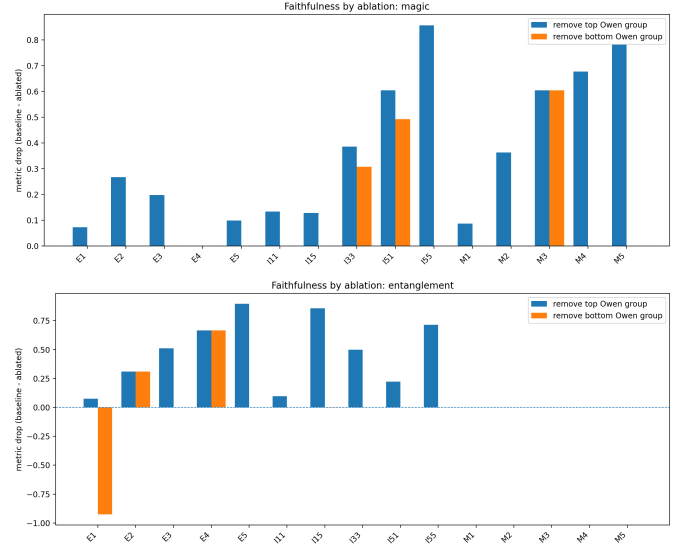


Fig. 2. Group-ablation support on the exact 15-circuit subset. The bars compare the value drop after removing the top-ranked Owen group and the bottom-ranked Owen group. Magic gives the clearest signal, while entanglement contains ties and near-zero baseline cases.

TABLE II
GROUP-ABLATION SUMMARY ON THE EXACT 15-CIRCUIT SUBSET.

Value function	Mean top drop	Mean bottom drop	Top wins
Magic	0.3502	0.0936	13/15
Entanglement	0.3233	0.0032	8/15

two non-winning cases correspond to one circuit with a single active group and one circuit with near-zero magic.

For the entanglement value function, the mean top-group drop is 0.3233, compared with 0.0032 for the bottom group, and the top group causes the larger drop in 8 out of 15 circuits. Out of seven non-winning cases, five have near-zero entanglement.

C. Monte Carlo Estimator Validation

The Monte Carlo Owen estimator was validated against exact Owen values on the 15-circuit subset using sampling fractions 0.30, 0.50, 0.70, and 0.90 with 10 random seeds per circuit and value function. The full validation plot and numerical table are reported in Appendix A-C. For magic, the normalized error decreases from 0.059 at 30% to 0.032 at 70% and remains 0.032 at 90%. For entanglement, the normalized error decreases from 0.061 at 30% to 0.041 at 70% and 0.037 at 90%. At 70%, the mean Spearman correlation is approximately 0.924 for both value functions. The full 35-circuit experiment therefore uses 70% sampling and averages five sampled runs per circuit and value function.

D. Estimated Owen Values on the Full 35-Circuit Benchmark

The full benchmark applies the sampled Owen estimator with sampling fraction 0.70 to all 35 circuits, averaging five sampled runs per circuit and value function. Figure 3 shows

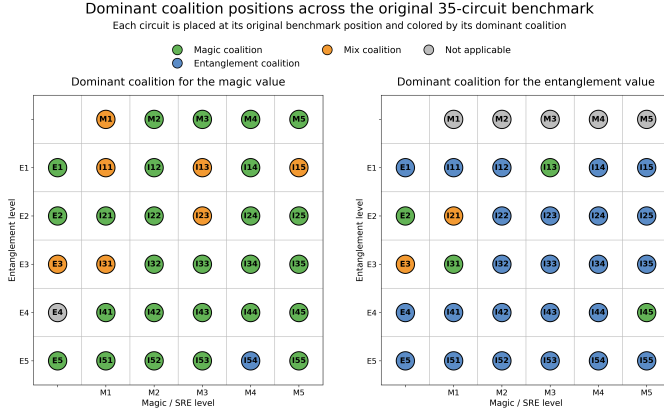


Fig. 3. Dominant Owen group on the full 35-circuit benchmark. Grey cells indicate circuits with numerically zero target value.

the dominant Owen group for each circuit. Circuits with numerically zero target value are shown in grey.

For the magic value, one circuit is not applicable because its SRE is zero. Among the remaining circuits, M is dominant in 26 circuits, X in 7 circuits, and E in 1 circuit. For the entanglement value, the five magic-axis circuits are not applicable because their entanglement value is zero. Among the remaining 30 circuits, E is dominant in 24 circuits, M in 4 circuits, and X in 2 circuits.

E. QNN Case Study

The QNN case study uses the two-qubit QNN described in Appendix A-D. Because the value function is one-shot classification accuracy, coalition values are stochastic. The E/M/X partition was evaluated with $K \in \{1, 8, 16, 32\}$ repeated value-function evaluations, and $K = 32$ is used for the final reported results. The full K comparison is reported in Appendix A-E. This stability check shows that the estimates are noisier for small K , but the main attribution pattern remains stable as K increases; nevertheless, the QNN results should be interpreted as one-shot accuracy explanations rather than as probability-based or multi-shot explanations.

For $K = 32$, the largest individual gate-level Owen value is assigned to gate 15 in the mixed group X . Additional positive contributions occur around gates 7, 16, and 17. The full gate-level plot is reported in Appendix A-F. A Shapley baseline recomputed with the same passive gates as the E/M/X Owen partition shows moderate but significant agreement with the QNN Owen gate ranking ($\rho = 0.525$, $p = 0.0445$; Appendix A-G).

The group-level totals in Fig. 4, reported as mean $\pm$ one standard deviation over five independent runs, are:

$$E = 0.0352 \pm 0.0120, M = 0.0133 \pm 0.0068, \\ X = 0.0670 \pm 0.0078$$

Thus, X has the largest group-level Owen total, followed by E and M .

Additional QNN partitions are reported in Appendix A-H.

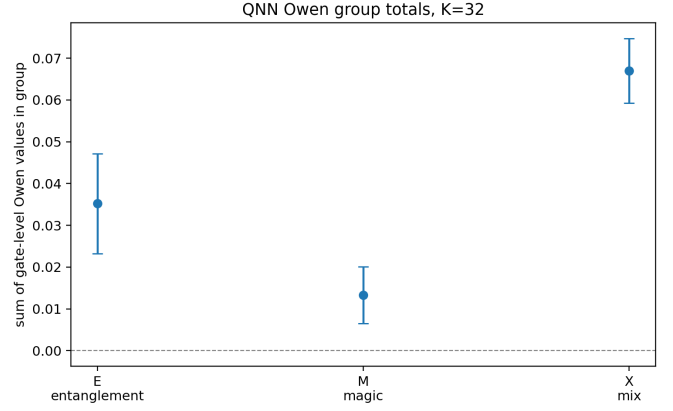


Fig. 4. Group-level QNN Owen values for the E/M/X partition with $K = 32$. The mixed group X receives the largest total attribution. Error bars show one standard deviation over five independent runs.

F. QSVM Case Study

The QSVM case study applies the E/M/X Owen explanation to the SVQX feature-map setup described in Appendix A-I. The value function is the QSVM test accuracy after retraining the classifier on the coalition kernel. Hadamard gates are treated as passive, while active gates are grouped into CNOT scaffold gates E , local phase rotations M , and cross-feature motif gates X ; exact gate indices are listed in Appendix A-I. Full-circuit accuracy checks are reported in Appendix A-J.

The mean gate-level Owen values over five datasets have their largest values on phase-encoding gates and intermediate feature-map gates, while final CNOT gates are often close to zero. The full gate-level plot is reported in Appendix A-L. When Shapley is recomputed with the same passive Hadamard gates as the Owen partition, the QSVM gate rankings show strong agreement with Owen values ($\rho = 0.843$ – 0.927 ; Appendix A-G).

The group-level Owen values in Fig. 5 show that M has the largest total attribution for all three values of r . The mixed group X is larger for $r = 2$ and $r = 3$ than for $r = 1$, while E remains smaller than M in all three settings. Additional feature-semantics and repetition-block partitions are reported in Appendix A-K; they show that the same QSVM case study can be interpreted in terms of feature encoding and feature-map depth.

VI. DISCUSSION

A. Interpretation of the Results

The benchmark results show that Owen values are sensitive to the value function being explained. On the exact 15-circuit subset, magic-axis circuits are assigned mainly to M or X , while entanglement-axis circuits assign E as top or co-top. The full 35-circuit benchmark shows the same overall pattern at larger scale: magic is mostly M -dominated, while entanglement is mostly E -dominated. This supports the idea that the attributions are not merely reflecting the manually chosen E/M/X labels, but the reduced-circuit game induced

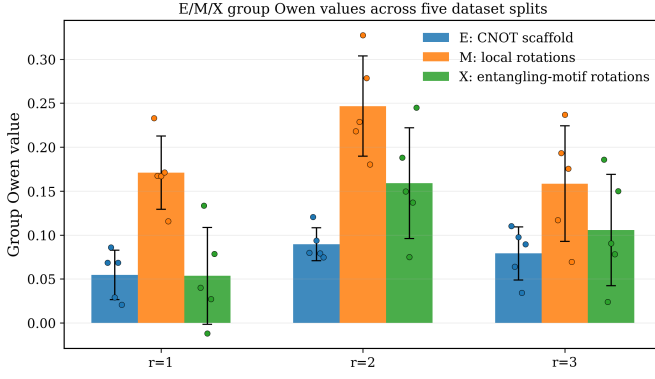


Fig. 5. QSVm group-level Owen values for the E/M/X partition. The local rotation group M is largest, while the mixed group X becomes more important for $r = 2$ and $r = 3$. Error bars show one standard deviation over five train/test splits.

by each value function. A label-aware random-partition control further supports this interpretation: E/M/X aligns magic attribution with M/X better than matched random partitions, and aligns entanglement attribution with E on circuits where entanglement is present (Appendix A-B).

The mixed and exceptional cases are also important. Some magic explanations involve X , and some entanglement explanations involve M or X . This shows that quantum resources are not always cleanly localized in one type of gate: local rotations can affect the states on which entangling gates act, and mixed gates can mediate local–entanglement effects. This is consistent with the broader view that entanglement and magic can interact in circuit dynamics [8].

The group-ablation experiment supports the Owen rankings, but also shows the limits of this test. The signal is clearer for magic than for entanglement. For entanglement, zero target values, tied attributions, and degenerate active groups make the top–bottom comparison less decisive. Overall, the ablation results support the rankings especially for magic, while showing that group removal is less informative when the target value is close to zero or when several groups interact.

The Monte Carlo validation shows that sampled Owen values can preserve the main ranking structure while reducing exact enumeration cost. The chosen sampling fraction gives a practical trade-off between runtime and accuracy for the full benchmark. However, this validation is still performed on small circuits where exact Owen values are available, so larger circuits may require additional estimator checks.

The QNN and QSVm case studies show that Owen values can add a module-level view to SVQX-style gate-level explanations. In the QNN case study, repeated coalition evaluations reduce the instability caused by the one-shot accuracy value function, and the largest group-level contribution comes from the mixed local–entanglement group. In the QSVm case study, the local phase-rotation group dominates across repetition depths, while the mixed group becomes more important for deeper feature maps. The locked-passive Shapley comparison shows that Owen values retain part of the gate-level Shapley

structure when both methods use the same active/passive convention. The agreement is moderate for the stochastic QNN case, but strong for the deterministic QSVm case studies. The group totals therefore provide a compact module-level interpretation on top of a broadly compatible gate-level attribution pattern. This is useful because quantum circuits are often designed as structured blocks rather than as unrelated individual gates [7].

B. Limitations and Future Work

The main limitation is the dependence on the chosen coalition structure. The label-aware random-partition control suggests that the E/M/X grouping is more resource-aligned than matched random partitions on the exact benchmark subset, but this does not remove the need to justify the partition. A meaningful partition can reveal useful module-level behaviour, whereas a poor partition may give weak or misleading explanations. Owen explanations should therefore always report the exact gate grouping, its motivation, and, when possible, partition-sensitivity checks. The experiments are also limited to small three-qubit benchmark circuits and controlled SVQX case studies, so larger circuits, noisy simulations, hardware runs, and more complex QML models may behave differently.

This thesis is a proof-of-concept study on small, exactly-solvable circuits. Future work should study automatic group construction, larger and noisier circuits, hardware-based evaluation, and test whether Owen values can support quantum architecture search by identifying useful functional modules. For the QNN case study, future work should also compare the one-shot accuracy value function with a more stable multi-shot or probability-based value function, because the present experiment reduces one-shot noise by repeated coalition evaluations but does not replace the value function itself.

VII. CONCLUSION

This thesis investigated whether Owen values can provide group-structured explanations for quantum circuits. For RQ1, the thesis showed that Owen values can be adapted to SVQX-style explanations by treating selected gates as players, grouping them into predefined functional modules, and evaluating each coalition by running the corresponding reduced circuit. The same framework can explain different value functions, including magic, entanglement, and classification accuracy, and can be computed exactly or approximated with Monte Carlo sampling.

For RQ2, the controlled benchmark showed that Owen values produce meaningful group-wise explanations. Magic was mostly attributed to local or mixed non-Clifford structure, while entanglement was mostly attributed to entangling gates. A label-aware random-partition control further supported this resource-specific interpretation on the exact benchmark subset. Group ablation supported these rankings, especially for magic, while also showing that near-zero values, ties, and group interactions can make ablation tests harder to interpret.

For RQ3, the QML case studies showed that Owen values can preserve Shapley-style gate-level structure when Shapley

is recomputed with the same passive gates as the Owen partition. The agreement was moderate for the stochastic QNN case and strong for the deterministic QSVM case studies. Across both case studies, Owen values added a module-level interpretation. Thus, Owen values do not replace Shapley values, but answer a complementary question: which predefined functional circuit groups matter for the chosen value function?

The main limitation is the dependence on the chosen coalition structure. Future work should study automatic group construction, larger and noisier circuits, hardware-based evaluation, and possible use in quantum architecture search. More interpretable quantum-circuit explanations can also have broader societal value: as quantum machine learning develops, methods that make model behaviour easier to inspect can support transparency, debugging, and more responsible use of quantum technologies.

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APPENDIX A

COALITION STRUCTURES AND ADDITIONAL EXPERIMENT RESULTS

This appendix reports the exact gate partitions used in the controlled 35-circuit benchmark, QNN case study, and QSVM case study, together with additional stability checks and supporting result tables. All gate indices are thesis-style 1-based indices.

A. Controlled Benchmark Coalition Structures

Tables III–IV list the gate sequences of the controlled benchmark circuits. Each entry has the form *index:gate(qubit)*. For CNOT gates, $q_a \rightarrow q_b$ denotes control qubit q_a and target qubit q_b .

Tables V–VI report the corresponding E/M/X coalition assignments using 1-based gate indices. Passive gates are kept in every reduced circuit but do not receive Owen attributions.

A dash indicates that no gate in that circuit belongs to the corresponding role; empty roles are omitted from the active partition during Owen-value computation.

B. Label-Aware Random-Partition Control

Because Owen values depend on the chosen partition, a random-partition control was used to test whether the resource-based E/M/X grouping carries information beyond group sizes alone. For each circuit in the exact benchmark subset, random partitions were generated by shuffling the same active gates into groups with the same sizes as the non-empty E/M/X groups. The original E/M/X partition itself was excluded from the random samples.

The control is label-aware: it does not only test whether a partition produces large group differences, but whether attribution is assigned to the physically expected resource groups. For a partition with group-level Owen totals Ψ_E, Ψ_M , and Ψ_X , the absolute attribution share of group G is defined as

$$s_G = \frac{|\Psi_G|}{|\Psi_E| + |\Psi_M| + |\Psi_X| + \epsilon}. \quad (20)$$

For the magic value function, the expected resource groups are M and X , so the resource-alignment score is

$$A_{\text{magic}} = s_M + s_X. \quad (21)$$

For the entanglement value function, the expected resource group is E , so

$$A_{\text{ent}} = s_E. \quad (22)$$

A second label-aware metric uses ablation of the expected groups. For magic, the expected group set is $M \cup X$; for entanglement, it is E . The normalized expected-group drop is

$$D_{\text{exp}} = \frac{v(N) - v(N \setminus G_{\text{exp}})}{|v(N)| + \epsilon}, \quad (23)$$

where $G_{\text{exp}} = M \cup X$ for magic and $G_{\text{exp}} = E$ for entanglement. For entanglement, circuits with zero entanglement value were excluded from this control, because there is no nonzero entanglement signal to attribute.

For each circuit and metric, the E/M/X value was compared with the matched random partitions using a percentile score. A percentile of 0.5 corresponds to random-level performance, while values above 0.5 indicate that E/M/X outperforms matched random partitions. Statistical significance was assessed using a one-sided Wilcoxon signed-rank test on the per-circuit percentiles against 0.5.

The results support the physical interpretation of the E/M/X grouping. For magic, the attribution is concentrated much more strongly in the expected M/X groups than in matched random partitions. The expected-group ablation also shows that removing M/X causes a larger normalized value drop than removing matched random groups. For entanglement, the same pattern is observed after excluding zero-entanglement circuits: attribution is more strongly aligned with E , and removing E causes a larger normalized drop than matched random removals. Thus, the E/M/X partition is not only manually motivated, but also shows stronger resource-specific alignment than random partitions with the same group sizes.

C. Estimator Validation Results

Table IX reports the numerical results used to validate the Monte Carlo Owen estimator on the exact 15-circuit subset. Figure 6 shows the corresponding error and runtime trends.

D. QNN Coalition Structures

Figure 7 shows the two-qubit QNN circuit and the gate indexing used in the Owen-value analysis. The main QNN result in Section V-E uses the resource-based E/M/X partition shown in Table X.

E. QNN Stability Across Repeated Evaluations

The main QNN result uses $K = 32$ repeated value-function evaluations per coalition. Figure 8 shows the corresponding stability check for $K \in \{1, 8, 16, 32\}$. The estimates are noisier for small K , especially for gates with small contributions, while the main attribution pattern remains stable as K increases.

F. QNN Gate-Level Owen Results

Figure 9 reports the final gate-level QNN Owen values for the E/M/X partition with $K = 32$.

G. Shapley–Owen Gate-Level Comparison

To quantify the relation between Owen values and standard gate-level Shapley/SVQX explanations, Shapley values were recomputed using the same active/passive convention as the corresponding Owen experiment. That is, gates that are passive in the Owen partition were kept fixed in every Shapley coalition, and only the Owen-active gates were treated as Shapley players. This gives a fairer comparison than directly comparing with the original SVQX-style Shapley baseline, because both methods explain the same reduced-circuit game and the same set of active gates.

Table XI reports the resulting rank and linear correlations. The QNN case shows moderate but statistically significant rank-level agreement. The Pearson correlation is close to

TABLE III
GATE DEFINITIONS FOR THE MAGIC-AXIS AND ENTANGLEMENT-AXIS BENCHMARK CIRCUITS.

Circuit	Gate sequence
M1	1 : ry(q_0), 2 : h(q_0), 3 : s(q_1), 4 : t(q_1), 5 : cx($q_1 \rightarrow q_0$), 6 : rx(q_1), 7 : rx(q_0), 8 : rz(q_1)
M2	1 : rz(q_0), 2 : rz(q_1), 3 : cx($q_1 \rightarrow q_2$), 4 : rz(q_1), 5 : rx(q_0), 6 : rx(q_2), 7 : rz(q_0), 8 : rz(q_1)
M3	1 : rx(q_0), 2 : rx(q_1), 3 : h(q_1), 4 : rz(q_0), 5 : rz(q_1), 6 : rx(q_2)
M4	1 : cx($q_0 \rightarrow q_1$), 2 : rx(q_2), 3 : ry(q_1), 4 : rx(q_1), 5 : rx(q_1), 6 : cx($q_0 \rightarrow q_2$), 7 : ry(q_0), 8 : rz(q_2)
M5	1 : ry(q_0), 2 : rz(q_1), 3 : cx($q_2 \rightarrow q_1$), 4 : rx(q_1), 5 : rz(q_0), 6 : rx(q_2), 7 : rz(q_2)
E1	1 : rx(q_0), 2 : rz(q_0), 3 : rx(q_0), 4 : s(q_0), 5 : h(q_1), 6 : cx($q_1 \rightarrow q_0$)
E2	1 : rz(q_0), 2 : rx(q_1), 3 : rx(q_1), 4 : ry(q_2), 5 : cx($q_1 \rightarrow q_0$), 6 : cx($q_1 \rightarrow q_2$)
E3	1 : rx(q_0), 2 : cx($q_1 \rightarrow q_0$), 3 : rz(q_2), 4 : s(q_2), 5 : t(q_2), 6 : rz(q_2), 7 : cx($q_0 \rightarrow q_1$)
E4	1 : h(q_0), 2 : h(q_0), 3 : z(q_0), 4 : h(q_1), 5 : x(q_1), 6 : s(q_1), 7 : cx($q_1 \rightarrow q_2$), 8 : h(q_0)
E5	1 : ry(q_0), 2 : t(q_1), 3 : rx(q_2), 4 : cx($q_0 \rightarrow q_2$), 5 : rz(q_1), 6 : cx($q_2 \rightarrow q_1$)

TABLE IV
GATE DEFINITIONS FOR THE INTERIOR BENCHMARK CIRCUITS.

Circuit	Gate sequence
I11	1 : t(q_0), 2 : rz(q_1), 3 : ry(q_2), 4 : cx($q_0 \rightarrow q_2$), 5 : cx($q_2 \rightarrow q_1$), 6 : h(q_1)
I12	1 : ry(q_0), 2 : rz(q_1), 3 : cx($q_0 \rightarrow q_1$), 4 : t(q_1), 5 : t(q_0), 6 : h(q_0), 7 : h(q_0), 8 : h(q_1), 9 : s(q_0), 10 : cx($q_1 \rightarrow q_0$), 11 : cx($q_0 \rightarrow q_1$)
I13	1 : s(q_0), 2 : h(q_0), 3 : t(q_1), 4 : rz(q_2), 5 : cx($q_1 \rightarrow q_0$), 6 : h(q_1), 7 : h(q_0), 8 : rx(q_1), 9 : s(q_0), 10 : t(q_2), 11 : rz(q_0), 12 : rz(q_0), 13 : rz(q_1), 14 : ry(q_2), 15 : cx($q_2 \rightarrow q_1$)
I14	1 : rx(q_0), 2 : ry(q_0), 3 : rz(q_1), 4 : cx($q_1 \rightarrow q_2$), 5 : rx(q_0), 6 : ry(q_0), 7 : rx(q_2), 8 : cx($q_0 \rightarrow q_2$), 9 : cx($q_2 \rightarrow q_1$), 10 : ry(q_1), 11 : cx($q_0 \rightarrow q_2$), 12 : rz(q_1), 13 : ry(q_2)
I15	1 : rz(q_0), 2 : ry(q_0), 3 : rx(q_1), 4 : cx($q_2 \rightarrow q_1$), 5 : cx($q_1 \rightarrow q_0$), 6 : cx($q_0 \rightarrow q_2$)
I21	1 : ry(q_0), 2 : cx($q_1 \rightarrow q_0$), 3 : cx($q_0 \rightarrow q_2$), 4 : rz(q_1), 5 : rz(q_0), 6 : rz(q_2), 7 : rx(q_0), 8 : cx($q_1 \rightarrow q_2$), 9 : ry(q_0), 10 : ry(q_1), 11 : ry(q_0), 12 : rx(q_0), 13 : ry(q_0), 14 : cx($q_2 \rightarrow q_0$)
I22	1 : rx(q_0), 2 : cx($q_1 \rightarrow q_2$), 3 : cx($q_1 \rightarrow q_2$), 4 : rz(q_1), 5 : rz(q_0), 6 : rz(q_0), 7 : rx(q_0), 8 : rx(q_2), 9 : rx(q_0), 10 : cx($q_2 \rightarrow q_0$)
I23	1 : z(q_0), 2 : t(q_1), 3 : h(q_2), 4 : t(q_0), 5 : h(q_0), 6 : cx($q_2 \rightarrow q_0$), 7 : t(q_0), 8 : x(q_0), 9 : h(q_0), 10 : t(q_0), 11 : y(q_2), 12 : cx($q_2 \rightarrow q_0$), 13 : s(q_2)
I24	1 : rx(q_0), 2 : rx(q_1), 3 : cx($q_2 \rightarrow q_0$), 4 : cx($q_0 \rightarrow q_2$), 5 : ry(q_1), 6 : h(q_2), 7 : ry(q_1), 8 : t(q_1), 9 : h(q_2), 10 : h(q_0), 11 : cx($q_1 \rightarrow q_2$), 12 : rz(q_1), 13 : s(q_1), 14 : h(q_0), 15 : rx(q_1)
I25	1 : s(q_0), 2 : h(q_0), 3 : cx($q_1 \rightarrow q_2$), 4 : rz(q_0), 5 : t(q_1), 6 : cx($q_0 \rightarrow q_1$), 7 : rx(q_2), 8 : rz(q_0), 9 : t(q_1), 10 : h(q_1), 11 : ry(q_2), 12 : s(q_1), 13 : h(q_1), 14 : rx(q_2), 15 : h(q_1), 16 : cx($q_1 \rightarrow q_2$), 17 : h(q_1)
I31	1 : ry(q_0), 2 : rz(q_1), 3 : rz(q_2), 4 : rx(q_2), 5 : cx($q_0 \rightarrow q_1$), 6 : rx(q_0), 7 : ry(q_0), 8 : rz(q_2), 9 : rx(q_2), 10 : cx($q_2 \rightarrow q_1$), 11 : rx(q_1), 12 : rz(q_0), 13 : cx($q_2 \rightarrow q_1$), 14 : rz(q_0), 15 : cx($q_2 \rightarrow q_1$)
I32	1 : rz(q_0), 2 : cx($q_1 \rightarrow q_0$), 3 : cx($q_1 \rightarrow q_2$), 4 : s(q_1), 5 : h(q_1), 6 : h(q_2), 7 : rz(q_2), 8 : rz(q_1), 9 : s(q_0), 10 : cx($q_1 \rightarrow q_2$), 11 : ry(q_2), 12 : rz(q_2), 13 : s(q_0), 14 : s(q_1), 15 : cx($q_2 \rightarrow q_0$), 16 : h(q_1), 17 : cx($q_2 \rightarrow q_0$), 18 : rx(q_1)
I33	1 : rx(q_0), 2 : cx($q_0 \rightarrow q_1$), 3 : ry(q_0), 4 : cx($q_0 \rightarrow q_1$), 5 : rz(q_2), 6 : ry(q_0), 7 : rz(q_1)
I34	1 : cx($q_0 \rightarrow q_1$), 2 : cx($q_2 \rightarrow q_1$), 3 : ry(q_2), 4 : ry(q_2), 5 : ry(q_2), 6 : rx(q_1), 7 : rx(q_2), 8 : cx($q_1 \rightarrow q_0$), 9 : cx($q_1 \rightarrow q_2$), 10 : rx(q_0), 11 : rz(q_1), 12 : rz(q_1), 13 : cx($q_1 \rightarrow q_0$), 14 : cx($q_2 \rightarrow q_0$)
I35	1 : cx($q_0 \rightarrow q_1$), 2 : rz(q_0), 3 : rz(q_0), 4 : ry(q_1), 5 : ry(q_1), 6 : rx(q_1), 7 : rz(q_1), 8 : rz(q_1), 9 : rz(q_0), 10 : rx(q_0), 11 : cx($q_1 \rightarrow q_2$), 12 : ry(q_1), 13 : cx($q_1 \rightarrow q_0$), 14 : ry(q_2), 15 : ry(q_2)
I41	1 : s(q_0), 2 : t(q_1), 3 : h(q_0), 4 : ry(q_0), 5 : ry(q_2), 6 : rz(q_2), 7 : rx(q_0), 8 : h(q_1), 9 : rx(q_2), 10 : rz(q_0), 11 : rx(q_2), 12 : ry(q_0), 13 : s(q_1), 14 : cx($q_0 \rightarrow q_2$), 15 : rx(q_2), 16 : h(q_0), 17 : rx(q_2), 18 : t(q_0), 19 : rx(q_1)
I42	1 : rx(q_0), 2 : cx($q_1 \rightarrow q_2$), 3 : rx(q_1), 4 : cx($q_0 \rightarrow q_1$), 5 : ry(q_1), 6 : rz(q_1), 7 : ry(q_0), 8 : ry(q_2), 9 : rz(q_2), 10 : rz(q_2), 11 : rz(q_0), 12 : rz(q_2), 13 : rz(q_1)
I43	1 : rx(q_0), 2 : cx($q_1 \rightarrow q_0$), 3 : ry(q_2), 4 : rz(q_2), 5 : ry(q_2), 6 : rz(q_1), 7 : rx(q_1), 8 : cx($q_2 \rightarrow q_1$), 9 : rz(q_0), 10 : ry(q_1), 11 : rx(q_0), 12 : rx(q_2), 13 : ry(q_0), 14 : cx($q_0 \rightarrow q_1$)
I44	1 : ry(q_0), 2 : cx($q_1 \rightarrow q_2$), 3 : rz(q_0), 4 : ry(q_2), 5 : rx(q_1), 6 : cx($q_1 \rightarrow q_0$), 7 : cx($q_0 \rightarrow q_1$), 8 : rz(q_2), 9 : cx($q_2 \rightarrow q_1$), 10 : ry(q_2), 11 : rz(q_1), 12 : ry(q_2), 13 : rx(q_1), 14 : rz(q_0)
I45	1 : s(q_0), 2 : h(q_1), 3 : s(q_2), 4 : h(q_0), 5 : cx($q_1 \rightarrow q_2$), 6 : h(q_2), 7 : rx(q_2), 8 : rz(q_0), 9 : ry(q_1), 10 : s(q_2), 11 : cx($q_2 \rightarrow q_0$), 12 : h(q_1), 13 : rx(q_0), 14 : cx($q_1 \rightarrow q_2$)
I51	1 : rx(q_0), 2 : rz(q_0), 3 : ry(q_1), 4 : rx(q_2), 5 : cx($q_1 \rightarrow q_2$), 6 : ry(q_1), 7 : ry(q_2)
I52	1 : cx($q_0 \rightarrow q_1$), 2 : rx(q_0), 3 : rx(q_2), 4 : cx($q_0 \rightarrow q_2$), 5 : cx($q_0 \rightarrow q_2$), 6 : ry(q_1), 7 : ry(q_2), 8 : ry(q_0), 9 : cx($q_1 \rightarrow q_0$), 10 : rx(q_0), 11 : rz(q_1), 12 : rx(q_1), 13 : rz(q_0), 14 : rx(q_0), 15 : cx($q_0 \rightarrow q_1$)
I53	1 : h(q_0), 2 : rx(q_0), 3 : s(q_1), 4 : cx($q_1 \rightarrow q_2$), 5 : s(q_2), 6 : cx($q_0 \rightarrow q_1$), 7 : ry(q_0), 8 : cx($q_0 \rightarrow q_1$), 9 : h(q_1), 10 : ry(q_2), 11 : rz(q_2), 12 : rx(q_2), 13 : t(q_0), 14 : cx($q_2 \rightarrow q_1$), 15 : rx(q_1), 16 : h(q_0), 17 : s(q_0), 18 : rx(q_0), 19 : rz(q_1)
I54	1 : ry(q_0), 2 : cx($q_0 \rightarrow q_1$), 3 : cx($q_2 \rightarrow q_0$), 4 : rx(q_0), 5 : cx($q_1 \rightarrow q_0$), 6 : rz(q_0), 7 : rz(q_2), 8 : rz(q_1), 9 : ry(q_2), 10 : rz(q_0), 11 : ry(q_2), 12 : cx($q_1 \rightarrow q_0$), 13 : cx($q_2 \rightarrow q_0$), 14 : ry(q_0), 15 : rx(q_0), 16 : cx($q_1 \rightarrow q_0$), 17 : ry(q_2), 18 : rz(q_0), 19 : rz(q_1)
I55	1 : rx(q_0), 2 : ry(q_1), 3 : rz(q_1), 4 : rx(q_2), 5 : rz(q_1), 6 : cx($q_0 \rightarrow q_1$), 7 : cx($q_1 \rightarrow q_0$), 8 : cx($q_2 \rightarrow q_0$)

zero, indicating that the exact attribution magnitudes are not linearly aligned in this stochastic one-shot setting. In contrast, the QSVM case studies show strong Spearman and Pearson correlations once the passive Hadamard gates are treated consistently. This supports the view that Owen values can retain useful gate-level Shapley-style structure, while still adding a group-conditioned module-level interpretation. Figures 10–13 show the corresponding gate-level comparisons.

H. Additional QNN Partitions

In addition to the resource-based E/M/X partition, three alternative QNN partitions were tested as a sensitivity analysis. These partitions are not the main focus of the QNN case study, but they show that Owen values can answer different module-level questions depending on the chosen coalition structure. For these partitions, gates $\{1, 3\}$ are passive, following the original SVQX QNN setup.

Table XIII reports the group-level Owen summaries for these additional partitions. The architecture partition assigns the largest total attribution to the trained classifier block

TABLE V
E/M/X COALITION STRUCTURES FOR THE MAGIC-AXIS AND ENTANGLEMENT-AXIS BENCHMARK CIRCUITS.

Circuit	Passive	E	M	X
M1	{2, 3}	{5}	{1, 7, 8}	{4, 6}
M2	–	{3}	{1, 5, 6, 7, 8}	{2, 4}
M3	{3}	–	{1, 2, 4, 5, 6}	–
M4	–	{1, 6}	{2, 3, 4, 5, 8}	{7}
M5	–	{3}	{1, 5, 6, 7}	{2, 4}
E1	{4, 5}	{6}	{1, 2, 3}	–
E2	–	{5, 6}	{1, 2, 3, 4}	–
E3	{4}	{2, 7}	{3, 5, 6}	{1}
E4	{1, 2, 3, 4, 5, 6, 8}	{7}	–	–
E5	–	{4, 6}	{1, 2}	{3, 5}

TABLE VI
E/M/X COALITION STRUCTURES FOR INTERIOR BENCHMARK CIRCUITS.

Circuit	Passive	E	M	X
I11	{6}	{4, 5}	{1, 2}	{3}
I12	{6, 7, 8, 9}	{3, 10, 11}	{1, 5}	{2, 4}
I13	{1, 2, 6, 7, 9}	{5, 15}	{3, 4, 8, 10, 11, 12, 13}	{14}
I14	–	{4, 8, 9, 11}	{1, 2, 5, 6, 12, 13}	{3, 7, 10}
I15	–	{4, 5, 6}	{1, 2}	{3}
I21	–	{2, 3, 8, 14}	{4, 5, 6, 7, 9, 10, 11, 12}	{1, 13}
I22	–	{2, 3, 10}	{1, 5, 6, 7, 8}	{4, 9}
I23	{1, 3, 5, 8, 9, 11, 13}	{6, 12}	{2, 4, 10}	{7}
I24	{6, 9, 10, 13, 14}	{3, 4, 11}	{1, 2, 5, 7, 8, 15}	{12}
I25	{1, 2, 10, 12, 13, 15, 17}	{3, 6, 16}	{4, 7, 8, 9, 11, 14}	{5}
I31	–	{5, 10, 13, 15}	{1, 2, 3, 4, 7, 8, 12, 14}	{6, 9, 11}
I32	{4, 5, 6, 9, 13, 14, 16}	{2, 3, 10, 15, 17}	{7, 8, 12, 18}	{1, 11}
I33	–	{2, 4}	{5, 6, 7}	{1, 3}
I34	–	{1, 2, 8, 9, 13, 14}	{4, 5, 6, 7, 10, 11}	{3, 12}
I35	–	{1, 11, 13}	{3, 4, 5, 6, 7, 8, 9, 10, 14, 15}	{2, 12}
I41	{1, 3, 8, 13, 16}	{14}	{2, 4, 5, 6, 7, 9, 10, 11, 12, 17, 18, 19}	{15}
I42	–	{2, 4}	{1, 6, 7, 8, 9, 10, 11, 12, 13}	{3, 5}
I43	–	{2, 8, 14}	{3, 4, 5, 6, 9, 10, 11, 12}	{1, 7, 13}
I44	–	{2, 6, 7, 9}	{1, 3, 4, 11, 12, 13, 14}	{5, 8, 10}
I45	{1, 2, 3, 4, 6, 10, 12}	{5, 11, 14}	{7, 8, 9, 13}	–
I51	–	{5}	{1, 2, 3, 7}	{4, 6}
I52	–	{1, 4, 5, 9, 15}	{6, 7, 11, 12, 13}	{2, 3, 8, 10, 14}
I53	{1, 3, 5, 9, 16, 17}	{4, 6, 8, 14}	{2, 10, 11, 12, 13, 18, 19}	{7, 15}
I54	–	{2, 3, 5, 12, 13, 16}	{7, 8, 9, 10, 11, 17, 18, 19}	{1, 4, 6, 14, 15}
I55	–	{6, 7, 8}	{1, 2, 3, 4}	{5}

TABLE VII
LABEL-AWARE RANDOM-PARTITION CONTROL ON THE EXACT BENCHMARK SUBSET. RANDOM PARTITIONS MATCH THE E/M/X GROUP SIZES. FOR ENTANGLEMENT, CIRCUITS WITH ZERO ENTANGLEMENT ARE EXCLUDED.

Value function	Metric	Expected group(s)	Valid circuits	E/M/X mean	Random mean	Mean percentile
Magic	Resource alignment	$M + X$	13	0.966	0.703	0.820
Magic	Expected-drop norm.	$M + X$	13	1.000	0.591	0.658
Entanglement	Resource alignment	E	9	0.505	0.321	0.709
Entanglement	Expected-drop norm.	E	9	1.000	-0.068	0.714

C . The feature-interaction partition highlights local feature encoding L and the classifier C , while the explicit cross-feature entangling motif group XFE is close to zero. The qubit-role partition assigns the largest total attribution to the measured-qubit path $Q0$, which is consistent with the fact that the model prediction is read from the first qubit.

I. QSVM Coalition Structure

Figure 14 shows one repetition block of the QSVM feature map and the gate indexing used in the Owen-value analysis. The QSVM feature map consists of $r \in \{1, 2, 3\}$ repetitions of a seven-gate block. For repetition $j \in \{1, \dots, r\}$, the gate indices in that block are shifted by $7(j - 1)$. The coalition structure is listed in Table XIV. The Hadamard gates are passive, the CNOT gates form the entanglement group, the

local phase gates form the magic group, and the cross-feature phase gate forms the mixed group.

J. Additional QSVM Results

Table XV reports the full-circuit QSVM accuracy over five train/test splits. This is used as a setup check before interpreting the Owen attributions.

Table XVI reports the group-level Owen values.

K. Additional QSVM Partitions

In addition to the resource-based E/M/X partition, two alternative QSVM partitions were tested as sensitivity analyses. These partitions are not used as the main QSVM explanation, but they show that Owen values can answer different module-level questions depending on the chosen coalition structure.

TABLE VIII

STATISTICAL SUMMARY OF THE LABEL-AWARE RANDOM-PARTITION CONTROL. PERCENTILES MEASURE HOW OFTEN THE E/M/X SCORE IS LARGER THAN THE MATCHED RANDOM PARTITIONS FOR THE SAME CIRCUIT AND VALUE FUNCTION. THE WILCOXON TEST IS ONE-SIDED AGAINST PERCENTILE 0.5.

Value function	Metric	Valid circuits	Mean percentile	Median percentile	Wilcoxon p
Magic	Resource alignment	13	0.820	0.800	1.22×10^{-4}
Magic	Expected-drop norm.	13	0.658	0.720	0.0239
Entanglement	Resource alignment	9	0.709	0.800	0.0137
Entanglement	Expected-drop norm.	9	0.714	0.750	0.0078

Monte Carlo Owen Estimator Validation
Fraction-based sampling; Pareto panels show normalized error vs runtime

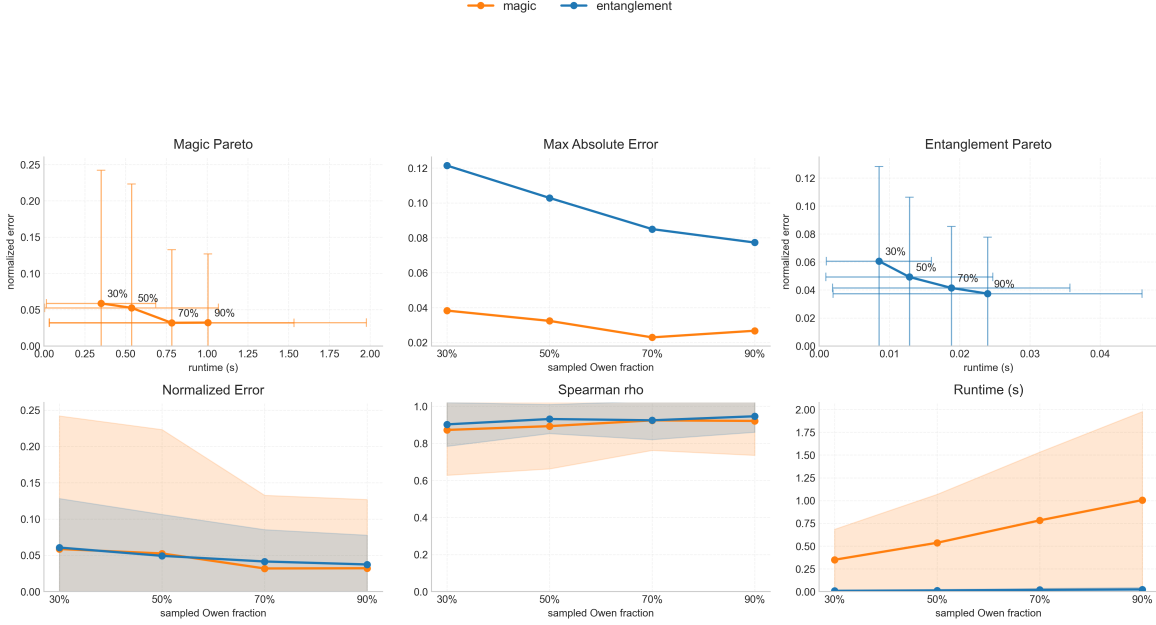


Fig. 6. Monte Carlo Owen estimator validation on the exact 15-circuit subset. Errors generally decrease as the sampling fraction increases, while runtime increases. Shaded regions show ± 1 standard deviation across circuits.

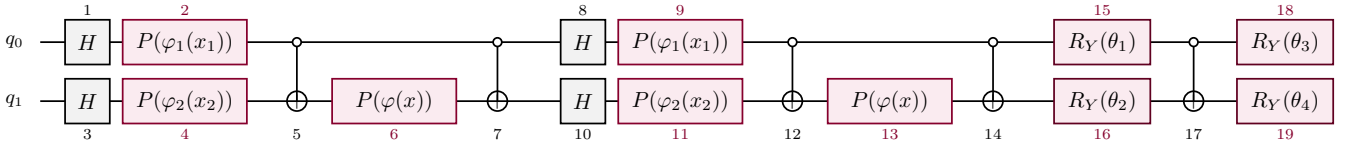


Fig. 7. Two-qubit QNN circuit used in the case study, redrawn from the SVQX experiment in [5]. Gate numbers correspond to the indexing used in the Owen-value analysis. The trained parameters are fixed at $\theta \approx (3.860, -1.070, -1.583, 0.860)$ during the explanation experiment.

TABLE IX

ESTIMATOR-VALIDATION RESULTS. RUNTIME IS THE MEAN WALL-CLOCK TIME PER SAMPLED OWEN RUN.

Property	Fraction	Norm. error	Spearman	Runtime
Magic	0.30	0.059	0.872	0.35s
Magic	0.50	0.053	0.893	0.54s
Magic	0.70	0.032	0.924	0.78s
Magic	0.90	0.032	0.921	1.01s
Ent.	0.30	0.061	0.902	0.009s
Ent.	0.50	0.049	0.931	0.013s
Ent.	0.70	0.041	0.924	0.019s
Ent.	0.90	0.037	0.946	0.024s

TABLE X

RESOURCE-BASED E/M/X PARTITION FOR THE QNN CASE STUDY.

Role	Gate indices
Passive	{1, 3, 8, 10}
E	{5, 7, 12, 14, 17}
M	{2, 4, 9, 11, 18, 19}
X	{6, 13, 15, 16}

The first alternative is a feature-semantics partition. For repetition block j , with offset $o = 7(j - 1)$, the Hadamard gates $\{1 + o, 3 + o\}$ are passive. The active groups are

$$F_1 = \{2+o\}, \quad F_2 = \{4+o\}, \quad F_{12} = \{5+o, 6+o, 7+o\}.$$

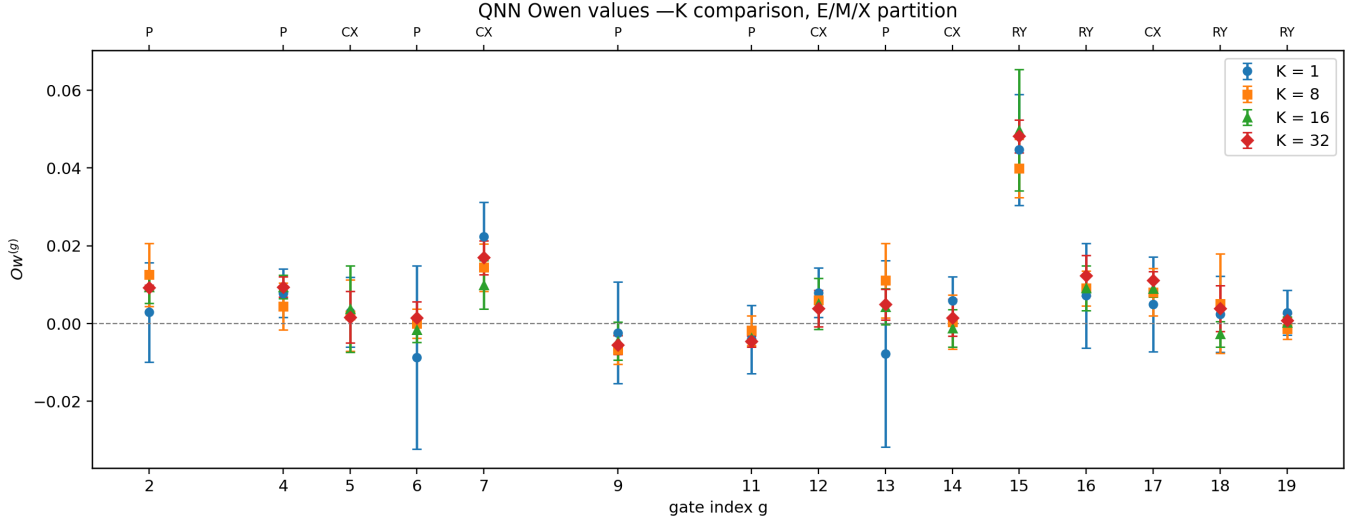


Fig. 8. Gate-level QNN Owen values under the E/M/X partition for different numbers of repeated value-function evaluations K . Error bars show variation over five independent runs.

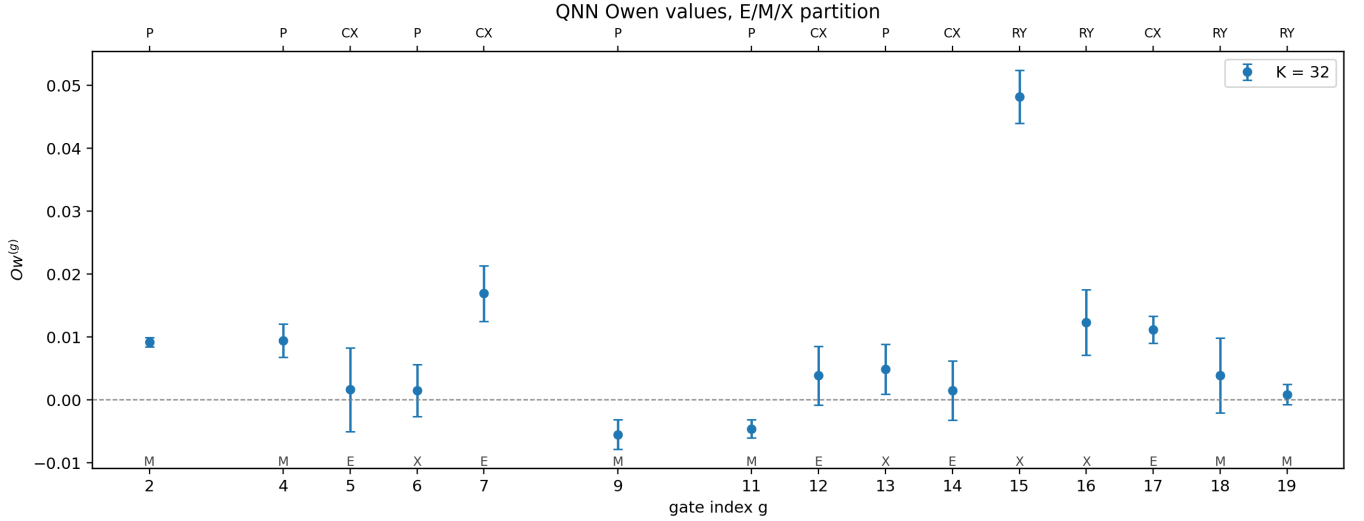


Fig. 9. Final gate-level QNN Owen values for the E/M/X partition with $K=32$, averaged over five independent runs. The largest attribution is assigned to gate 15 in the mixed group X.

Here, F_1 corresponds to the local phase encoding of feature x_1 , F_2 to the local phase encoding of feature x_2 , and F_{12} to the two-feature interaction motif.

The second alternative is a repetition-block partition. For each repetition block j , the active group is

$$B_j = \{2 + o, 4 + o, 5 + o, 6 + o, 7 + o\},$$

with the same passive Hadamard gates $\{1 + o, 3 + o\}$. This partition asks which feature-map repetition contributes most to the QSVM accuracy.

Table XVII summarizes the resulting group-level Owen values and absolute attribution shares. The feature-semantics partition assigns the largest share to F_2 for all repetition depths, suggesting that the local encoding of x_2 is consistently

important. However, for $r=2$ and $r=3$, the shares are distributed across F_1 , F_2 , and F_{12} , indicating that both single-feature encodings and the pairwise feature interaction contribute. The repetition-block partition shows that the first feature-map repetition receives the largest attribution. For $r=3$, the second block also contributes substantially, while the third block is smaller.

L. QSVM Gate-Level Owen Results

Figure 15 reports the mean gate-level QSVM Owen values with E/M/X partition over five datasets for $r \in \{1, 2, 3\}$.

TABLE XI
GATE-LEVEL COMPARISON BETWEEN OWEN VALUES AND LOCKED-PASSIVE SHAPLEY VALUES. THE SHAPLEY BASELINE USES THE SAME PASSIVE GATES AS THE CORRESPONDING E/M/X OWEN PARTITION.

Case	Active gates	Spearman ρ	Spearman p	Pearson r	Pearson p
QNN	15	0.525	0.0445	0.022	0.9369
QSVM $r = 1$	5	0.900	0.0374	0.979	0.0037
QSVM $r = 2$	10	0.927	1.12×10^{-4}	0.960	1.10×10^{-5}
QSVM $r = 3$	15	0.843	7.97×10^{-5}	0.890	8.61×10^{-6}

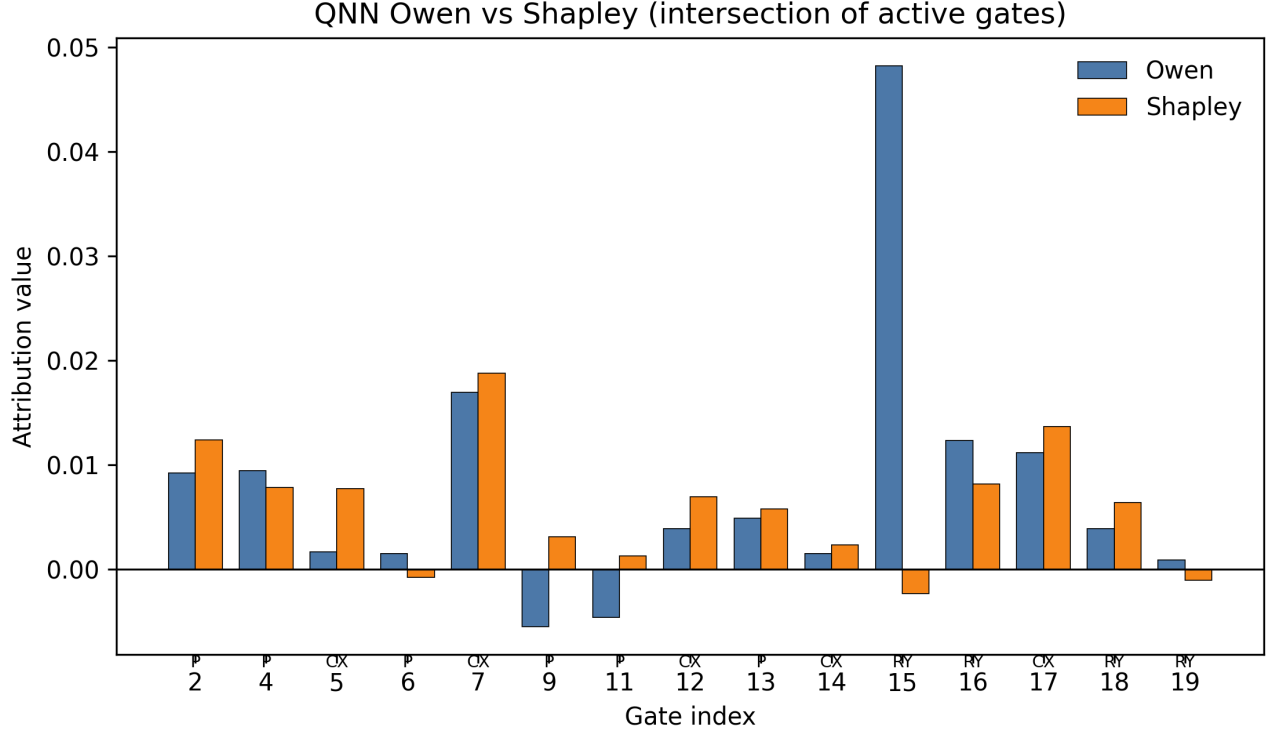


Fig. 10. Gate-level comparison between QNN Owen values under the E/M/X partition and locked-passive Shapley values computed with the same passive gates.

TABLE XII
ADDITIONAL QNN COALITION STRUCTURES. GATES $\{1, 3\}$ ARE PASSIVE.

Partition	Group	Gate indices
Architecture	B_1 feature-map block 1	$\{2, 4, 5, 6, 7\}$
	B_2 feature-map block 2	$\{8, 9, 10, 11, 12, 13, 14\}$
	C trainable classifier	$\{15, 16, 17, 18, 19\}$
Feature interaction	L local feature encoding	$\{2, 4, 8, 9, 10, 11\}$
	XFE cross-feature entangling motifs	$\{5, 6, 7, 12, 13, 14\}$
	C trainable classifier	$\{15, 16, 17, 18, 19\}$
Qubit role	$Q0$ measured-qubit path	$\{2, 8, 9, 15, 18\}$
	$Q1$ hidden-qubit path	$\{4, 6, 10, 11, 13, 16\}$
	$2Q$ two-qubit couplers	$\{5, 7, 12, 14, 17\}$
	$Q1F$ final hidden-qubit rotation	$\{19\}$

TABLE XIII
ADDITIONAL QNN OWEN SUMMARIES FOR ALTERNATIVE PARTITIONS. GROUP TOTALS ARE MEAN $\pm$ ONE STANDARD DEVIATION OVER FIVE INDEPENDENT RUNS AT $K = 32$.

Partition	Group totals	Main interpretation
Architecture	$B_1 = 0.0323 \pm 0.0130$; $B_2 = 0.0281 \pm 0.0119$; $C = 0.0432 \pm 0.0007$	Classifier block is largest.
Feature interaction	$L = 0.0627 \pm 0.0101$; XFE = 0.0013 ± 0.0113 ; $C = 0.0479 \pm 0.0122$	Local encoding and classifier dominate.
Qubit role	$Q0 = 0.0594 \pm 0.0143$; $Q1 = 0.0321 \pm 0.0095$; $2Q = 0.0310 \pm 0.0049$; $Q1F = -0.0011 \pm 0.0051$	Measured-qubit path is largest.

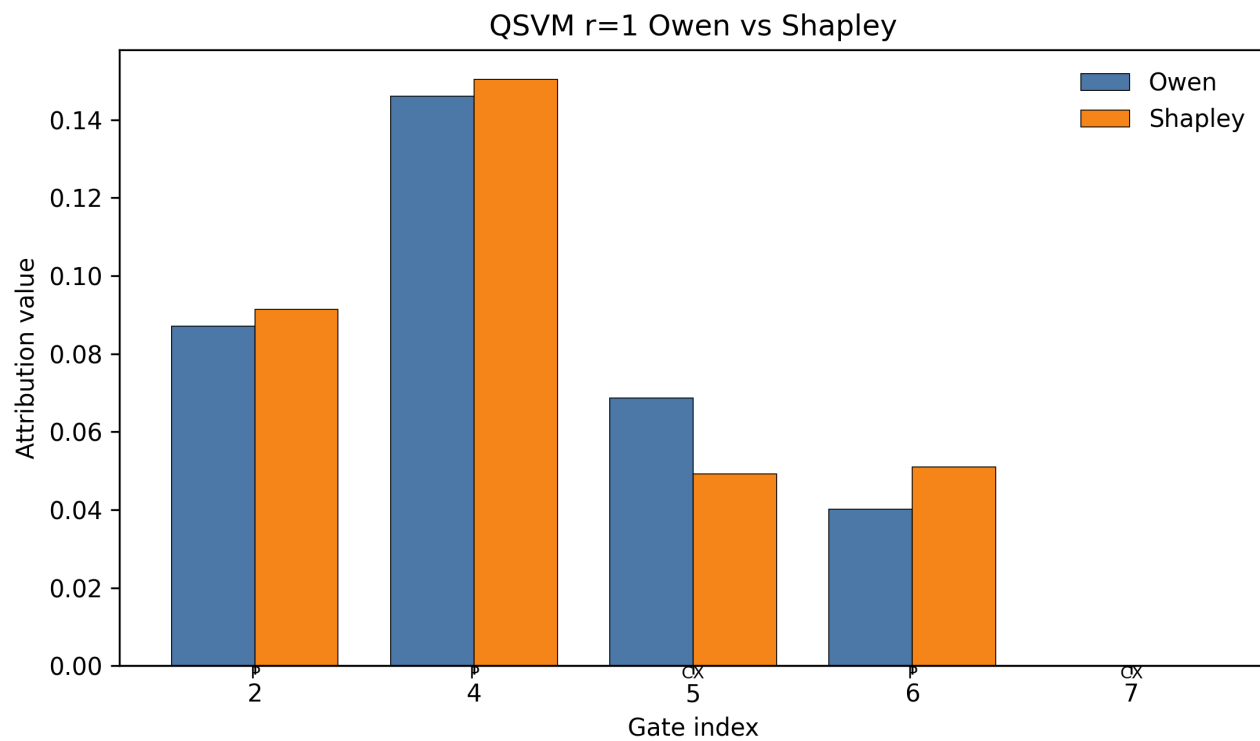


Fig. 11. Gate-level comparison between QSVM Owen values and locked-passive Shapley values for $r = 1$. Hadamard gates are passive in both calculations.

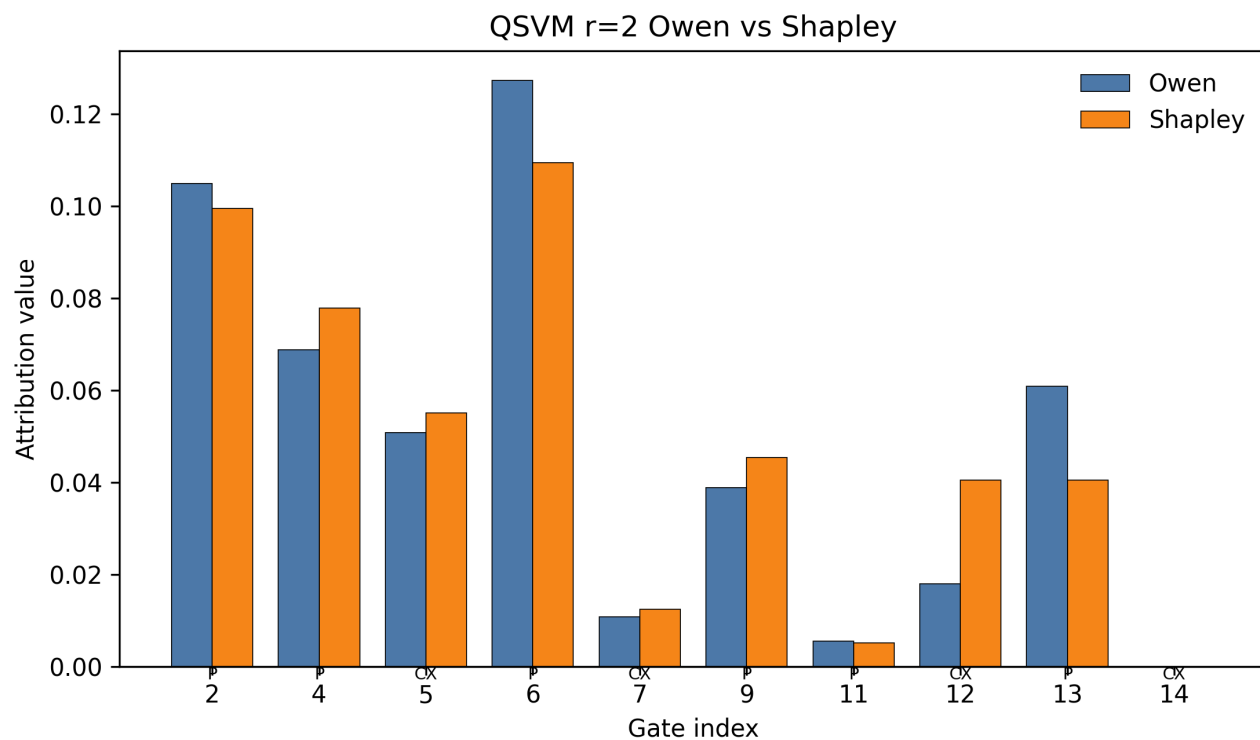


Fig. 12. Gate-level comparison between QSVM Owen values and locked-passive Shapley values for $r = 2$. Hadamard gates are passive in both calculations.

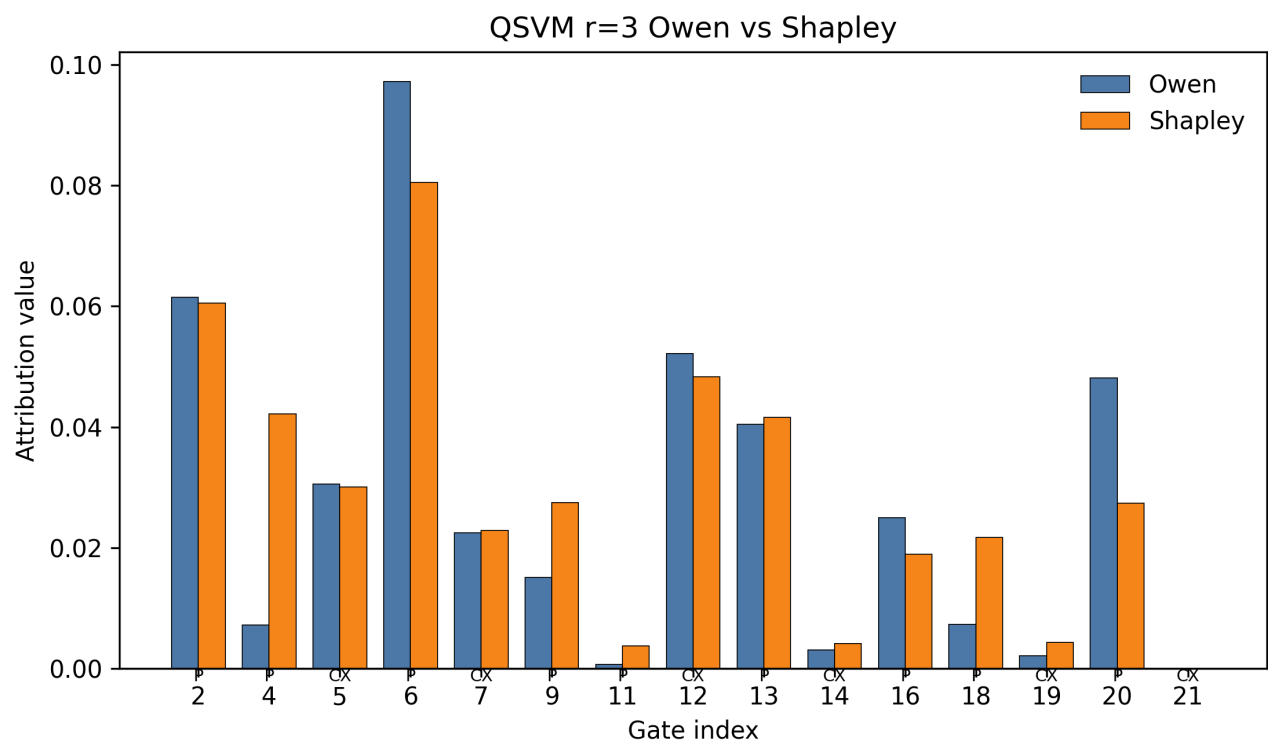


Fig. 13. Gate-level comparison between QSVM Owen values and locked-passive Shapley values for $r = 3$. Hadamard gates are passive in both calculations.

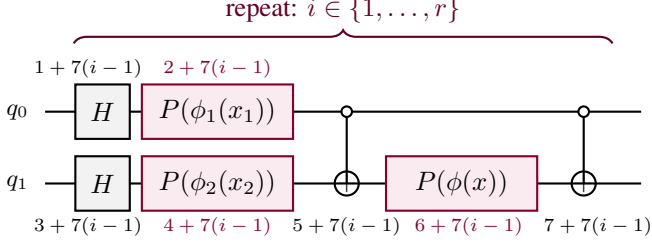


Fig. 14. One repetition block of the two-qubit QSVM feature map, redrawn from the SVQX experiment in [5]. The full feature map repeats this block $r \in \{1, 2, 3\}$ times, and gate numbers correspond to the Owen-value analysis.

TABLE XIV
QSVM E/M/X PARTITION FOR REPETITION j .

Role	Gate indices in repetition j
Passive	$\{1 + 7(j-1), 3 + 7(j-1)\}$
E	$\{5 + 7(j-1), 7 + 7(j-1)\}$
M	$\{2 + 7(j-1), 4 + 7(j-1)\}$
X	$\{6 + 7(j-1)\}$

TABLE XV
QSVM FULL-CIRCUIT ACCURACY OVER FIVE TRAIN/TEST SPLITS.

Repetitions	Mean	Std.	Range
$r = 1$	0.779	0.085	0.684–0.887
$r = 2$	0.995	0.006	0.986–1.000
$r = 3$	0.843	0.070	0.781–0.925

TABLE XVI
MEAN QSVM GROUP-LEVEL OWEN VALUES OVER FIVE TRAIN/TEST SPLITS. VALUES ARE SHOWN AS MEAN $\pm$ ONE STANDARD DEVIATION.

Repetitions	E	M	X
$r = 1$	0.0546 ± 0.0282	0.1709 ± 0.0417	0.0535 ± 0.0552
$r = 2$	0.0896 ± 0.0188	0.2467 ± 0.0571	0.1590 ± 0.0630
$r = 3$	0.0791 ± 0.0302	0.1584 ± 0.0658	0.1057 ± 0.0634

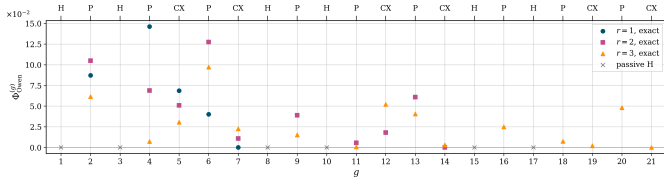


Fig. 15. Mean gate-level QSVM Owen values over five datasets for $r \in \{1, 2, 3\}$. High values repeatedly appear on phase-encoding gates and intermediate feature-map gates.

TABLE XVII
QSVN OWEN SUMMARIES FOR ALTERNATIVE MEANINGFUL PARTITIONS. GROUP VALUES ARE MEAN $\pm$ ONE STANDARD DEVIATION OVER FIVE
TRAIN/TEST SPLITS. SHARES ARE MEAN ABSOLUTE ATTRIBUTION SHARES.

Partition	r	Group values	Main interpretation
Feature-semantics	1	$F_1 = 0.0609 \pm 0.0357$ (0.201); $F_2 = 0.1399 \pm 0.0222$ (0.545); $F_{12} = 0.0782 \pm 0.0577$ (0.255)	F_2 is largest.
Feature-semantics	2	$F_1 = 0.1683 \pm 0.0294$ (0.340); $F_2 = 0.1868 \pm 0.0519$ (0.377); $F_{12} = 0.1402 \pm 0.0519$ (0.283)	Shared attribution, with F_2 slightly largest.
Feature-semantics	3	$F_1 = 0.1088 \pm 0.0504$ (0.314); $F_2 = 0.1308 \pm 0.0424$ (0.393); $F_{12} = 0.1035 \pm 0.0512$ (0.294)	Shared attribution, with F_2 largest.
Repetition blocks	1	$B_1 = 0.2790 \pm 0.0760$ (1.000)	Degenerate one-block partition.
Repetition blocks	2	$B_1 = 0.3871 \pm 0.0374$ (0.782); $B_2 = 0.1081 \pm 0.0388$ (0.218)	First repetition dominates.
Repetition blocks	3	$B_1 = 0.1683 \pm 0.0251$ (0.494); $B_2 = 0.1148 \pm 0.0367$ (0.331); $B_3 = 0.0602 \pm 0.0326$ (0.175)	Earlier repetitions contribute more.

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